

SMART WARRANTY PORTAL

1. Problem Statement

Traditional warranty management methods such as paper bills, manual records, and physical documents make it difficult for users to store, track, and manage product warranty information. Users may lose important warranty details or forget the warranty expiry date.

The Smart Warranty Portal is developed to provide a centralized digital platform where users can register products, store warranty details, view warranty status, and receive alerts when a warranty is expired or approaching its expiry date.

2. Objectives

The main objectives of the Smart Warranty Portal are:

To provide an easy-to-use platform for user registration and login.

To allow users to register their products digitally.

To store product and warranty information in a MySQL database.

To display warranty details in an organized manner.

To automatically identify Active, Expiring Soon, and Expired warranties.

To provide warranty alerts when a warranty is expired or nearing expiry.

To reduce manual paperwork and make warranty management easier.

To provide a user-friendly web interface.

3. scope

The scope of the Smart Warranty Portal includes:

User registration and login management.

Digital product registration.

Storage of product and warranty information.

Viewing registered products and warranty details.

Automatic warranty status calculation.

Warranty expiry and upcoming-expiry alerts.

Dashboard with product and warranty statistics.

Integration of React frontend with Spring Boot backend.

Database management using MySQL.

Deployment as a cloud-based web application.

4. Expected Outcome

The expected outcome of the project is a simple, centralized, and efficient warranty management system.

After implementation, users will be able to:

Register → Login → Register Product → View Warranty → Check Status → Receive Expiry Alert

The system will reduce manual warranty tracking, help users avoid missing warranty expiry dates, and provide quick access to product warranty information through a web-based application.